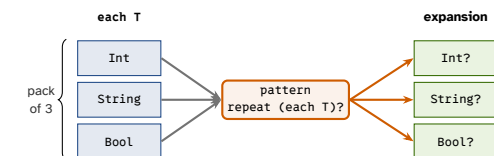


In one line: A **parameter pack** (5.9) lets one generic declaration take *any number* of type parameters: `<each T>` declares the pack, `repeat pattern` expands the pattern once per element, and each `x` names the current element — no more 1..10 overloads. Around it, every release since added small syntax that deletes boilerplate: expression `if/switch`, trailing commas, raw identifiers, `weak let`, module selectors, some `P?`, `@diagnose`, `await` in `defer`.

How packs work

- `func f<each T>(_ v: repeat each T): T` is a **type pack**, `v` a **value pack**; a call `f(1, "a", true)` binds `T = {Int, String, Bool}`. Constraints apply per element: `<each T: Equatable>`.
- `repeat P` is a **pack expansion**: `P` must mention a pack with `each`; it becomes a comma list — `repeat [each T] → [Int], [String], [Bool]`. Values: `(repeat each v)` builds a tuple (SE-0399).
- **Same shape**: two packs in one expansion (`repeat (each L, each R)`) must have the same length — the compiler infers and enforces it.
- **No count, no subscript**: you cannot index a pack. You expand it, or (6.0, SE-0408) iterate it: `for x in repeat each v`, one element per iteration, *lazily* — `break` stops evaluating the rest. In 5.9: a local generic `func + repeat try f(each v)`.
- **Variadic types** (SE-0398, 5.9): `struct Zip<each S: Sequence>`; at most **one** pack per type; store it as a tuple (`repeat each S`).
- Where you meet them: SwiftUI's `ViewBuilder` (no 10-child limit), zip-style APIs over any arity, typed tuple helpers.

One pattern, stamped per element



same shape: `repeat (each L, each R)`



length 3 vs 2 → compile error; `for (a, b) in walks the pairs`

Example — packs

```
func tuplefy<each T>(_ v: repeat each T) -> (repeat each T) {
    (repeat each v) // 5.9: tuple expansion
}
let t = tuplefy(1, "a", true) // (Int, String, Bool)

func allEqual<each E: Equatable>(_ l: (repeat each E),
    _ r: (repeat each E)) -> Bool {
    for (a, b) in repeat (each l, each r) { // 6.0 pack iteration
        guard a == b else { return false } // rest never evaluated
    }
    return true
}
allEqual((1, "a"), (1, "b")) // false; E = Int, String
struct Zip<each S: Sequence> { let seqs: (repeat each S) }
```

Example — the new syntax

```
let tint: Color = if ok { .green } else { .red } // 5.9
let p = Point(x: 1, y: 2,) // 6.1
@Test func `rejects an empty email`() { // 6.2
    enum Shade { case `50`, `100` }; let s = Shade.`50` // 6.2
    final class Tag: Sendable { weak let owner: Owner? } // 6.3
    let x = ModuleA::getValue() // 6.3
    weak var delegate: any CellDelegate? // 6.4
    @diagnose(DeprecatedDeclaration, as: warning,
        reason: "legacy bridge until 3.0") // 6.4
    func bridge() { oldAPI() }
    func upload() async throws {
        let h = try await open()
        defer { await withTaskCancellationShield { await h.close() } }
        try await h.send(payload) // 6.4
    }
```

(Owner: a Sendable class; CellDelegate: AnyObject-bound.)

Feature	Swift	SE	The rule that bites
if/switch expression	5.9	0380	only as return value, assignment, let/var init — not inside a larger expression; else mandatory; one expression per branch; branches type-checked independently (<code>if c { 0 } else { 1.0 }</code>) fails — annotate the type; a <code>throw/fatalError</code> branch is fine
trailing commas	6.1	0439	<code>in ()</code> , <code>[]</code> , <code>< ></code> lists: args, params, tuples, subscripts, capture lists, generics, interpolation. Not in <code>if/guard</code> conditions, inheritance clauses, where, <code>case a, b</code>
raw identifiers	6.2	0451	any characters between backticks except backtick, backslash, newline; not all whitespace, not only operator characters; use sites need the backticks too
weak let	6.3	0481	still becomes <code>nil</code> when the object dies, but cannot be reassigned ⇒ a <code>final</code> class with it can be <code>Sendable</code>
module selectors <code>M::x</code>	6.3	0491	pick the module when two imports export the same name (also on members: <code>s.Foundation::data(using::)</code>); lookup starts at module top level, so locals/generic params are unreachable; never on a new declaration's name
some <code>P?</code> / any <code>P?</code>	6.4	0521	<code>= (some P)? / (any P)?</code> — the optional wraps the opaque/existential, no parentheses needed
@diagnose	6.4	0522	<code>@diagnose(Group, as: error warning ignored, reason: "...")</code> on one declaration (<code>func</code> , <code>type</code> , <code>extension</code> , <code>import...</code>); order-sensitive; module flag <code>-suppress-warnings</code> still wins
await in defer	6.4	0493	only in an <code>async</code> context; awaited on every scope exit (a hidden suspension point); cancellation is not suppressed — wrap in <code>withTaskCancellationShield</code> (SE-0504)
access-level imports	6.0	0409	<code>internal import X</code> keeps <code>X</code> out of your public API (error if a public signature uses it); a bare <code>import</code> stays public in modes 5 and 6 unless <code>InternalImportsByDefault</code>
package access	5.9	0386	visible to other modules built with the same <code>-package-name</code> (SwiftPM sets it); between public and internal; no subclassing outside the module

Traps

- ! “Just use `v.count`” — a pack has no count or index; iterate (6.0) or expand.
- ! some `P?` is an *optional of an opaque type*, not an opaque optional: the underlying type is still fixed.
- ! `defer { await ... }` in a cancelled task: `Task.sleep`, `network` calls inside it fail at once — shield the cleanup.
- ! `weak let` is **6.3**, not 6.2 (proposal header).

Remember

“each names it, repeat stamps it, for walks it.”